2. Continuous Distributions

This chapter introduces random variables and continuous distributions.

First, some basic concepts of probability are reviewed.

- Two interpretations of the probability of an event.
  - *Frequency interpretation*: the probability is a property of the outcome of the experiment. It is the proportion of the repeated experiments that result in the outcome (for example, tossing a coin, 60% of similar regions contain oil)
  - *Subjective interpretation*: a statement of the beliefs of someone concerning the chance of an outcome (for example, one's belief is that there is a 60% chance that a particular region contains oil, the probability of you having a car accident in the coming year).
  - The mathematics of probability are the same in either case.

- The ***sample space, S***, is the set of all possible outcomes. Roll of a die, S={1,2,3,4,5,6}.

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- Rules for operations of events:
  - Commutative law: $E \cup F = F \cup E$ and $E \cap F = F \cap E$
  - Associative law: $(E \cup F) \cup \mathrm{G} = \mathrm{E} \cup (\mathrm{F} \cup G)$ and $(E \cap F) \cap \mathrm{G} = \mathrm{E} \cap (\mathrm{F} \cap G)$
  - Distributive law: $(E \cup F) \cap \mathrm{G} = (\mathrm{E} \cap G) \cup (\mathrm{F} \cap G)$
    and $(E \cap F) \cup \mathrm{G} = (\mathrm{E} \cup G) \cap (\mathrm{F} \cup G)$

- ***Axioms of probability***
  - ***Axiom 1:*** $0 \leq P(E) \leq 1$
  - ***Axiom 2:*** $P(S) = 1$
  - ***Axiom 3:*** $P(\bigcup_{i=1}^{n} E_i) = \sum_{i=1}^{n} P(E_i)$ for mutually exclusive events, i.e. $E_i \cap E_j = \emptyset$

- Propositions:
  - $P(E^c) = 1 - P(E)$
  - $P(E \cup F) = P(E) + P(F) - P(E \cap F)$

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- Conditional probability:
  - $P(E|F) = \frac{P(E \cap F)}{P(F)}$

- ***Bayes' formula***
  - Since $\mathrm{E} = (E \cap F) \cup (E \cap F^c)$ with $E \cap F$ and $E \cap F^c$ being mutually exclusive, $\mathrm{P}(\mathrm{E}) = P(E|F)P(F) + P(E|F^c)P(F^c)$
  - In general, for mutually exclusive events $F_1, F_2, \ldots, F_n$ such that $\bigcup_{i=1}^{n} F_i = S$

    $$\mathrm{P}(F_j|E) = \frac{\mathrm{P}(F_j \cap E)}{P(E)}$$

    $$\mathrm{P}(F_j|E) = \frac{\mathrm{P}(E|F_j)\mathrm{P}(F_j)}{\sum_{i=1}^{n} \mathrm{P}(E|F_j)\mathrm{P}(F_j)}$$ Known as Bayes' formula.

- Independent events
  - E and F are independent if $P(E|F) = P(E)$ => $P(E \cap F) = P(E)P(F)$

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2.1 Random Variables

2.1.1 A Definition

A **random variable** is a numerical quantity of interest determined by the result of a random experiment.

Example 1: Find the probability that the sum of two fair dice is at most 4.

The outcome of the experiment, i.e. rolling two dice, is (*i,j*) for $i \in \{1,2,\dots,6\}$ *and* $j \in \{1,2,\dots,6\}$ where *i* is the result of die 1 and *j* the one for die 2.

The sample space, S, contains 36 events, each with probability 1/36.

Letting X be the random variable of the sum, X can take values in {2,3,…,12}.

We want $\mathrm{P}(\mathrm{X} \leq 4) = P(\mathrm{A})$ where A is the set of events with a sum of 2, 3 or 4.

Since A={(1,1), (1,2), (1,3), (2,1), (2,2), (3,1)} , we get $\mathrm{P}(\mathrm{X} \leq 4) = \frac{6}{36} = \frac{1}{6}$.

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Example 2: A device will function normally if at least one of its two electronic components is not defective. Suppose a new electronic component has a 30% chance of being defective. A device is fitted with two (independent) new electronic components. Let the random variable X be the number of functioning (acceptable) electronic components.

a) Find the probability that X will take each possible value.

The 4 possible outcomes are (d,d), (d,a), (a,d) and (a,a).

P{(d,d)} = (0.3)(0.3) =0.09; P{(d,a)}=P{(a,d)} = (0.3)(0.7)=0.21; P{(a,a)} = 0.49

P(X=0)=0.09; P(X=1)=0.21+0.21=0.42; P(X=2)=0.49

b) Find the probability that the device is functioning normally.

Let I_A denote the indicator variable for event A. Using, $I_A = \begin{cases} 1 & \text{if } X = 1 \text{ or } 2 \\ 0 & \text{if } X = 0 \end{cases}$

$$P(I_A = 1) = P(X = 1) + P(X = 2) = 0.91$$
$$= 1 - P(X = 0) = 0.91$$

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2.1.2 Types of Random Variables

A **discrete random variable** is one that can take values in

- A finite sequence $x_1, x_2, \ldots, x_n$; or
- An infinite sequence $x_1, x_2, \ldots$

} Countable sets

A **continuous random variable** is one that can take on a continuum of possible values.

For example, the lifetime of a device, the amount of rain in a month, the inflation rate, …

Example: State whether the following random variables are discrete (finite or infinite) or continuous.

i. The final grade in a course for a student chosen at random.

ii. The time a randomly selected student takes to run a kilometre.

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2.1.3 Distribution function and Probability density function

The **cumulative distribution function (CDF)** or distribution function, $\boldsymbol{F(x)}$, of the random variable X is the probability that X takes a value less are equal to x. That is

$$F(x) = \mathrm{P}(\mathrm{X} \leq x)$$

We write $\mathrm{X} \sim F$ to indicate that F is the distribution function of X.

All probabilities about X can be found using its CDF.

For example, we can use Axiom 3 to find $\mathrm{P}(a < \mathrm{X} \leq b)$.

$$\mathrm{P}(\mathrm{X} \leq b) = \mathrm{P}(\mathrm{X} \leq a) + \mathrm{P}(a < \mathrm{X} \leq b)$$

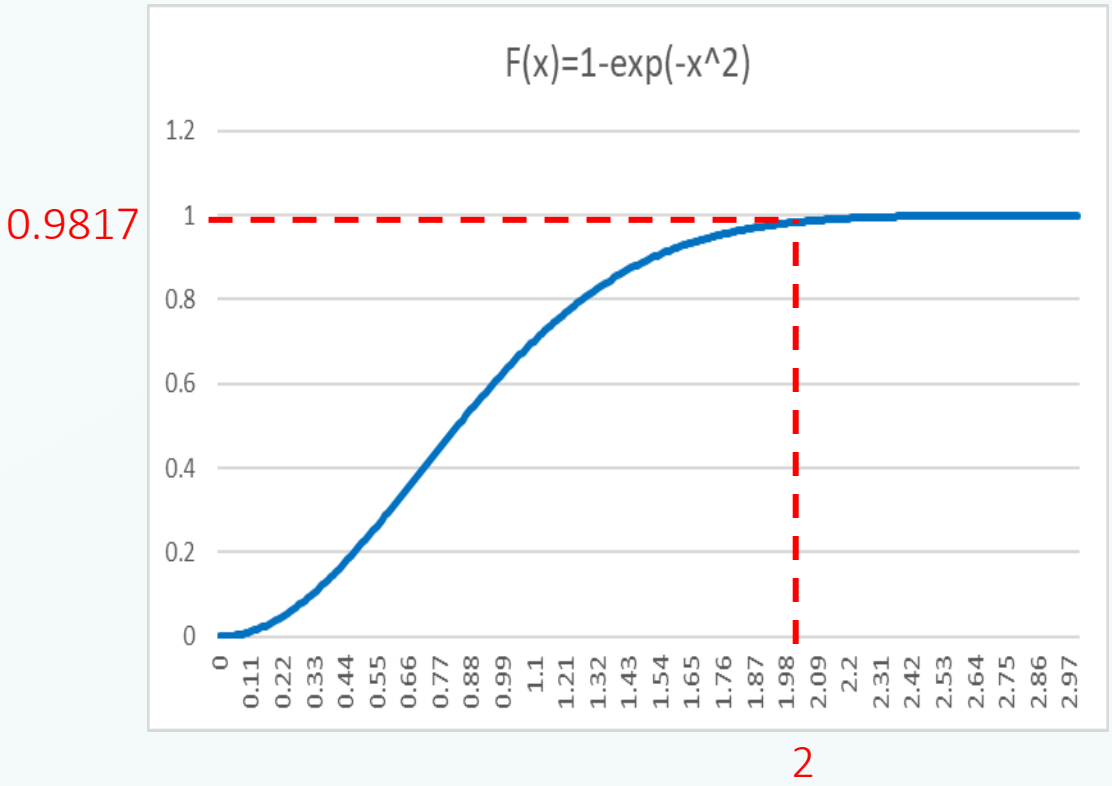
Mutually exclusive events

$$\mathrm{P}(a < \mathrm{X} \leq b) = \mathrm{P}(\mathrm{X} \leq b) - \mathrm{P}(\mathrm{X} \leq a) = F(b) - F(a)$$

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Example: Suppose $\mathrm{X} \sim F$ with $F(x) = \begin{cases} 0 & x \leq 0 \\ 1 - \exp(-x^2) & x > 0 \end{cases}$. Find $\mathrm{P}(\mathrm{X} > 2)$.

$$\mathrm{P}(\mathrm{X} > 2) = 1 - \mathrm{P}(\mathrm{X} \leq 2) = 1 - (1 - \exp(-4)) = 0.0183$$



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The **probability mass function**, $\mathrm{p}(x)$, of the *discrete* random variable X is the probability that X takes a value equal to x. That is

$$p(x) = \mathrm{P}(\mathrm{X} = x)$$

Note that $p(x) > 0$ if x is one of the possible values $x_1, x_2, \ldots$ and 0 otherwise.

The CDF of X in terms of $p(x)$ is

$$F(a) = \sum_{x \leq a} p(x)$$

It is a right-continuous step function.

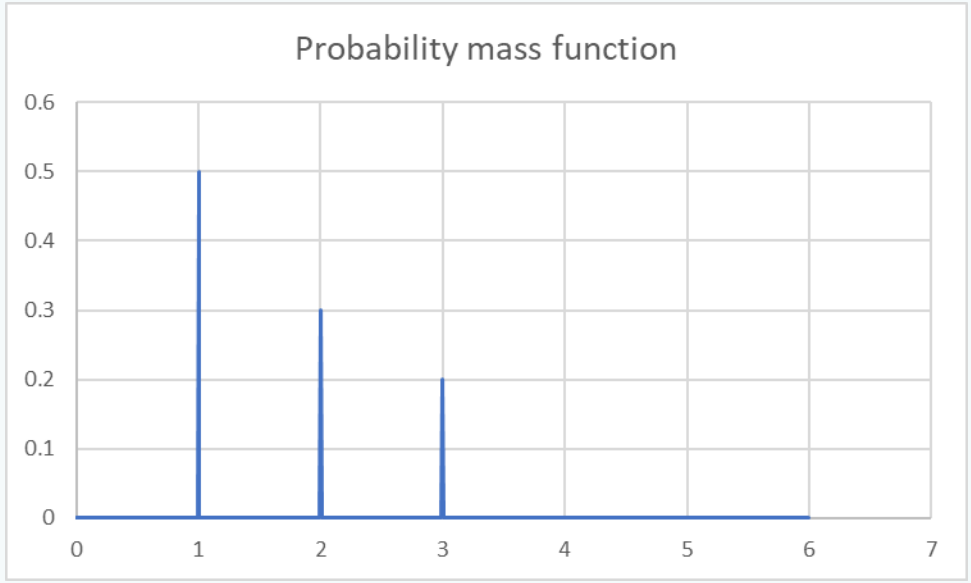
It is constant in $[x_{i-1}, x_i)$ and jumps by $p(x_i)$ at x_i.

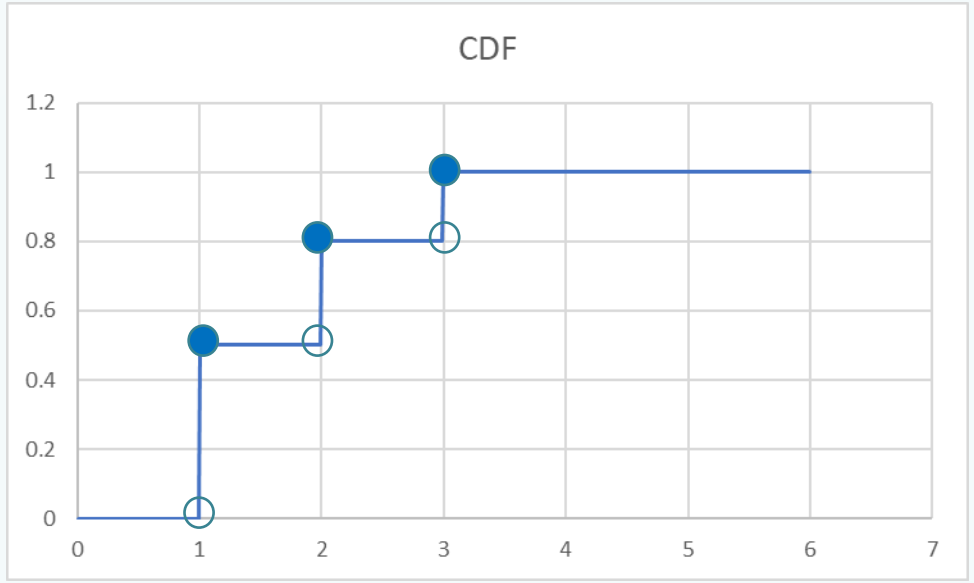
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Example: Let $\mathbf{X}$ take values 1, 2 or 3 with probabilities

$$p(1) = 0.5, p(2) = 0.3, p(3) = 0.2$$

Graph its probability mass function and its CDF.





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The **probability density function**, $f_X(x)$, or simply $f(x)$, of the *continuous* random variable $\mathbf{X}$ is a nonnegative function defined for all real x and having the property that, for any A,

$$P\{X \in A\} = \int_A f(x)\,dx$$

Note that from Axiom 2, $\int_{-\infty}^{\infty} f(x)\,dx = 1.$ P(S)=1

All probabilities about $\mathbf{X}$ can be found using its pdf.
For example:

- Letting $A = [a, \mathrm{b}]$, $P\{X \in A\} = \mathrm{P}(a \le X \le b) = \int_a^b f(x)\,dx$
- Letting $A = \{a\}$, $P\{X \in A\} = \mathrm{P}(\mathrm{X} = a) = \int_a^a f(x)\,dx = 0$ Let b=a

The probability that X, a continuous random variable, takes any particular value is 0.

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The pdf is the derivative of the CDF.

Let $A = [-\infty, a]$, then

$$F(a) = P\{X \in A\} = \int_{-\infty}^{a} f(x)\,dx$$

Taking the derivative with respect to a, we get

$$\frac{d}{da}F(a) = \frac{d}{da}\int_{-\infty}^{a} f(x)\,dx = f(a)$$

Note: It is commonly expressed as $\frac{d}{dx}F(x) = f(x)$

Intuitive interpretation of the pdf.

The probability that X will take a value in a small interval of length ε near the value a is approximately $\varepsilon\, f(a)$.

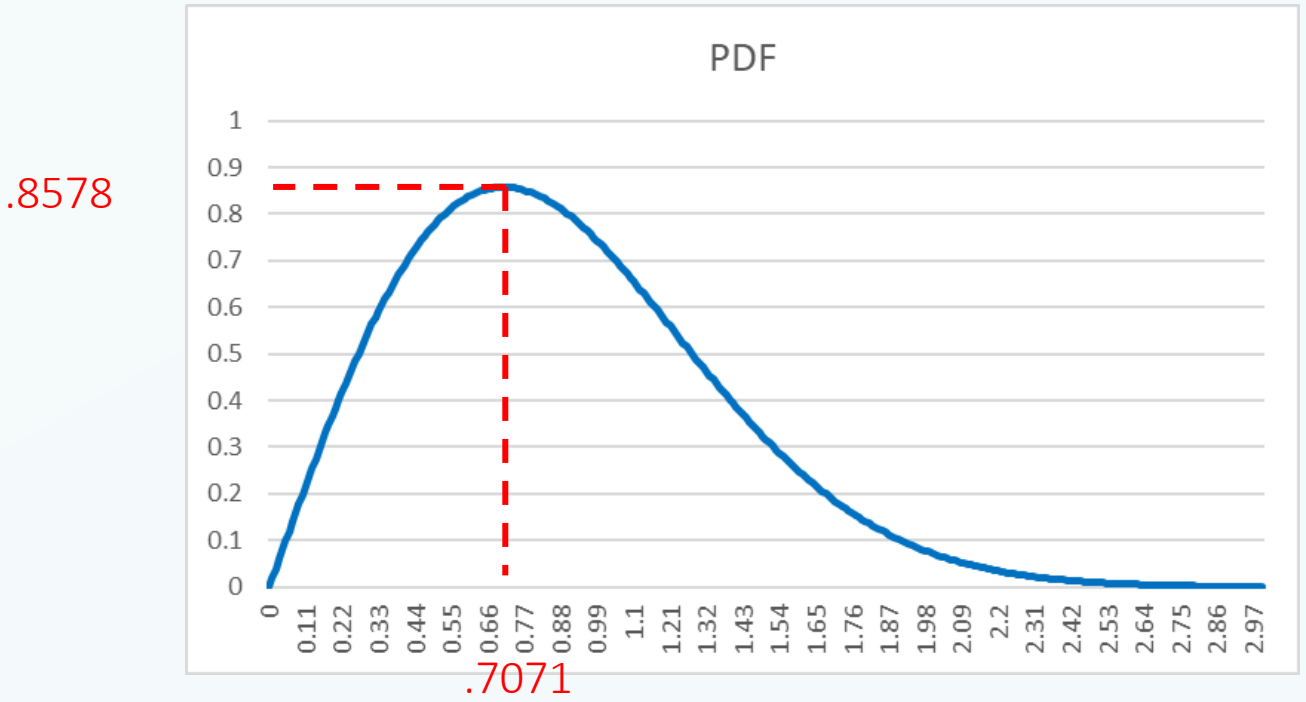
$$\mathrm{P}\left(a - \tfrac{\varepsilon}{2} \le X \le a + \tfrac{\varepsilon}{2}\right) = \int_{a-\frac{\varepsilon}{2}}^{a+\frac{\varepsilon}{2}} f(x)\,dx \approx \varepsilon\, f(a).$$

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Example: Suppose $X \sim F$ with $F(x) = \begin{cases} 0 & x \leq 0 \\ 1 - \exp(-x^2) & x > 0 \end{cases}$.

Find the pdf of X, $f(x)$.

$$f(x) = \frac{d}{dx} F(x) = \begin{cases} & x \leq 0 \\ & x > 0 \end{cases}$$



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Example:

Let $\mathbf{X}$ be a continuous random variable on [0,2]. Its cumulative distribution function is

$$F(x) = \begin{cases} 0 & x \leq 0 \\ C(2x^2 - \frac{2}{3}x^3) & 0 < x < 2 \\ 1 & 2 \leq x \end{cases}$$

a) Find the value of the constant C.

b) Find the probability density function of $\mathbf{X}$.

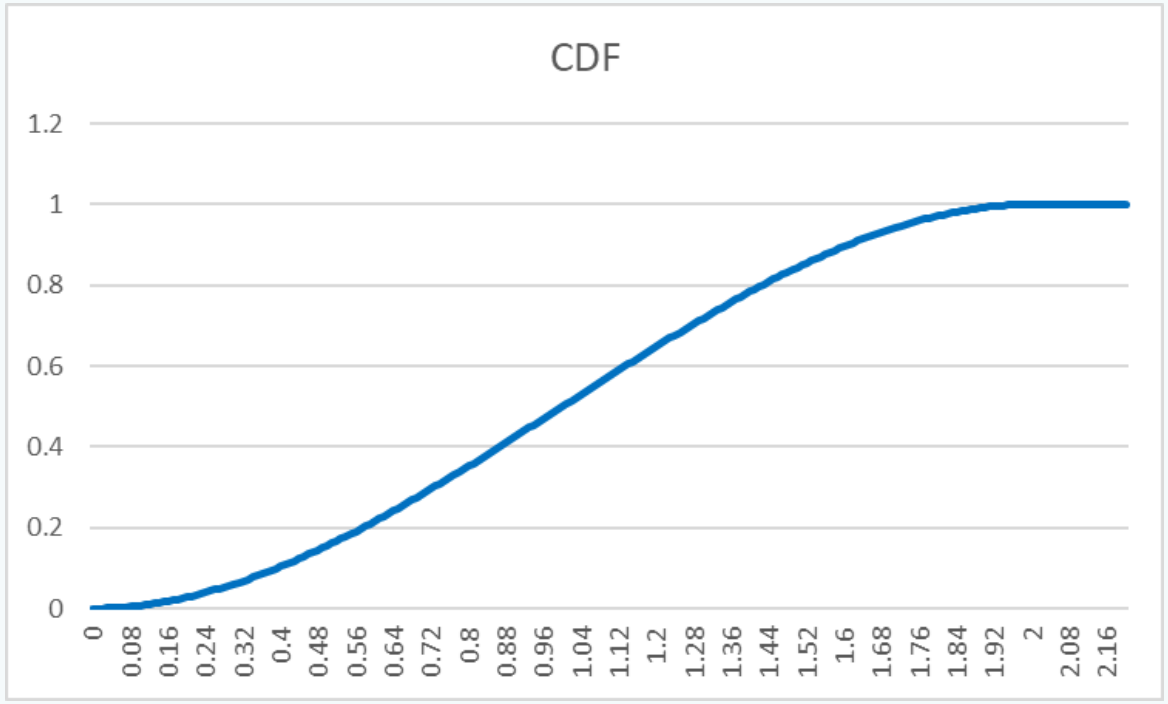
c) Find $\mathbf{P(0.5 < X < 1.5).}$

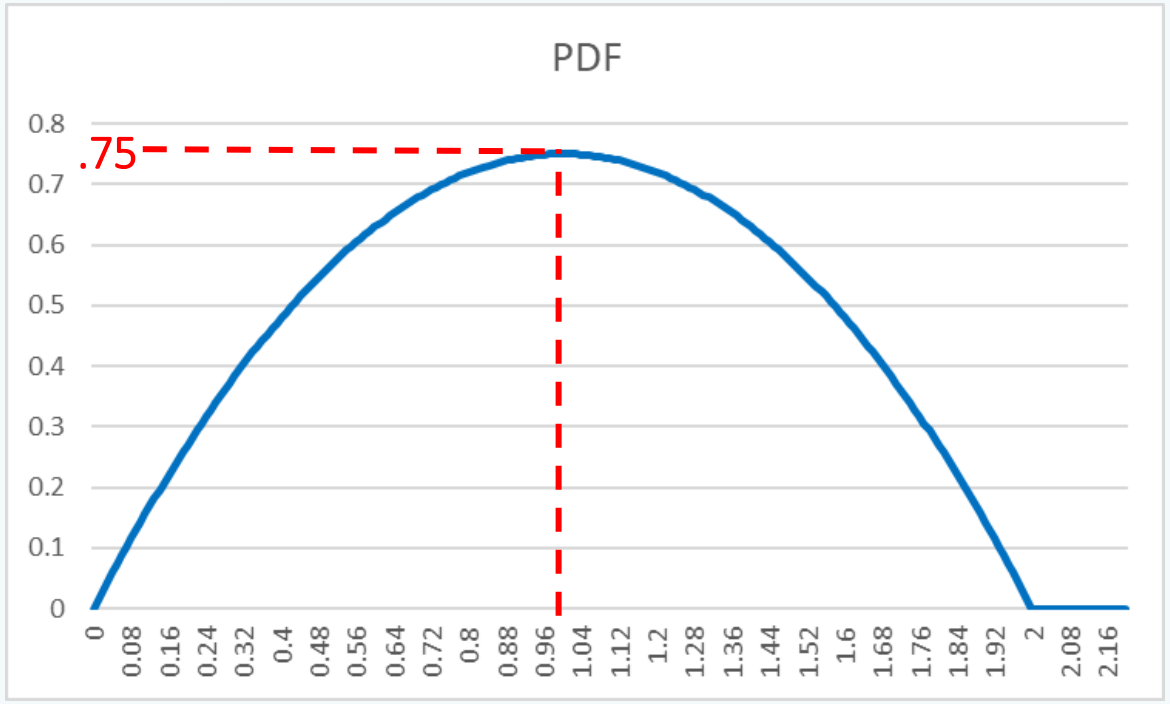
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Example:

Let $\mathbf{X}$ be a continuous random variable on [0,2]. Its cumulative distribution function is

$$F(x) = \begin{cases} 0 & x \le 0 \\ C(2x^2 - \frac{2}{3}x^3) & 0 < x < 2 \\ 1 & 2 \le x \end{cases} \Rightarrow f(x) = \begin{cases} 1.5x - .75x^2 & 0 \le x \le 2 \\ 0 & \text{otherwise} \end{cases}$$





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Exercise:

Suppose the time in minutes to complete a certain game is a random variable with pdf

$$f(x) = \begin{cases} 0 & 0 < x < 30 \\ C/x^2 & 30<\text{x} \end{cases}.$$

What is the probability that exactly 2 games out of 6 independent games will be completed within two hours?

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2.1.4 Jointly distributed random variables

For some experiments, we are interested in more than one random variable.

For example, one might be interested in the relationship between the average number of cigarettes smoked daily and the age at which a person is diagnosed with cancer.

Another example is the relationship between the shear strength and the diameter of a spot weld in a sheet steel specimen.

The relationship between two random variables can be specified by the **joint cumulative distribution function** , $\boldsymbol{F(x, y)}$, of X and Y

$$\boldsymbol{F(x, y) = P\{X \leq x, Y \leq y\}.}$$

All probabilities about X and Y can be obtained from their joint CDF.

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The **marginal CDF's** of X and Y can be obtained from their joint CDF.

$$F_X(x) = P(X \leq x) = P(X \leq x, Y < \infty) = F(x, \infty)$$

and

$$F_Y(y) = P(Y \leq y) = P(X < \infty, Y \leq y) = F(\infty, y).$$

Remarks:

-Knowledge of the joint CDF of X and Y allows us to find their corresponding marginal CDF's. $F(x, y) \Rightarrow F(x) \text{ and } F(y)$

-The reverse is not true. Knowledge of the marginal CDF's does not allow one to find the joint CDF. $F(x) \text{ and } F(y) \nRightarrow F(x, y)$

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If X and Y are both discrete, their joint probability mass function is defined as

$$p(x_i, y_j) = P\{X = x_i, \mathrm{Y} = y_j\}.$$

Using Axiom 3,

$$P\{X = x_i\} = P\left(\bigcup_j \{X = x_i, \mathrm{Y} = y_j\}\right) = \sum_j P\{X = x_i, \mathrm{Y} = y_j\} = \sum_j p(x_i, y_j)$$

Of course, we have $P\{\mathrm{Y} = y_j\} = \sum_i p(x_i, y_j)$.

X and Y are **jointly continuous** if there exists a function $\mathrm{f}(x, y)$, called the joint probability density function, satisfying

$$\mathrm{P}\{(\mathrm{X}, \mathrm{Y}) \in C\} = \int \int_{(x,y)\in C} \mathrm{f}(x, y)\, dx\, dy$$

for every C.

Letting $C = \{(x, y): x \in A, y \in B\}$,

$$\mathrm{P}\{X \in A, Y \in B\} = \int_B \int_A \mathrm{f}(x, y)\, dx\, dy = \int_A \int_B \mathrm{f}(x, y)\, dy\, dx$$

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Since $\mathrm{P}\{X \in (-\infty, a), Y \in (-\infty, b)\} = F(a,b) = \int_{-\infty}^{b} \int_{-\infty}^{a} \mathrm{f}(x,y)\, dx\, dy$,

differentiation with respect to a and b gives

$$\frac{\partial^2}{\partial a \partial b} F(a,b) = \mathrm{f}(a,b)$$

Equivalently, $\frac{\partial^2}{\partial x \partial y} F(x,y) = \mathrm{f}(x,y)$.

Intuitive interpretation of the joint pdf.

The probability that X will take a value in a small interval of length da near the value a and Y will take a value in a small interval of length db near the value b, is approximately $f(a,b)\, da\, db$.

$$\mathrm{P}(a \le X \le a + da, b \le Y \le b + db) =$$

$$= \int_{b}^{b+db} \int_{a}^{a+da} f(x,y)\, dx\, dy \approx f(a,b)\, da\, db.$$

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Since $\mathrm{P}\{X \in (a, a+da), Y \in (-\infty, \infty)\} = \int_a^{a+da} \int_{-\infty}^{\infty} f_{X,Y}(x,y)\, dy\, dx$

$$= \mathrm{P}\{X \in (a, a+da)\}$$

$$= \int_a^{a+da} f_X(x)\ dx$$

$\forall\ f_{X,Y}, f_X$

It follows that the marginal probability density function of X is

$$f_X(x) = \int_{-\infty}^{\infty} \mathrm{f}(x,y)\ dy$$

Similarly, $f_Y(y) = \int_{-\infty}^{\infty} \mathrm{f}(x,y)\ dx$.

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Example: The joint pdf of X and Y is

$$f(x,y) = \begin{cases} 2e^{-x}e^{-2y} & 0 < x, 0 < y \\ 0 & \text{otherwise} \end{cases}$$

Find **a)** $\boldsymbol{P(X > 1, Y < 1)}$; b) $P(X < Y)$; c) $P(X > a)$

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Example: The joint pdf of X and Y is

$$f(x,y) = \begin{cases} 2e^{-x}e^{-2y} & 0 < x, 0 < y \\ 0 & \text{otherwise} \end{cases}$$

Find a) $P(X > 1, Y < 1)$; **b)** $\boldsymbol{P(X < Y)}$; c) $P(X > a)$

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Example: The joint pdf of X and Y is

$$f(x,y) = \begin{cases} 2e^{-x}e^{-2y} & 0 < x, 0 < y \\ 0 & \text{otherwise} \end{cases}$$

Find a) $P(X > 1, Y < 1)$; b) $P(X < Y)$; **c)** $\boldsymbol{P(X > a)}$

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2.1.5 Independent random variables

X and Y are independent if for any two sets A, B,

$$P\{X \in A, Y \in B\} = P\{X \in A\}P\{Y \in B\}.$$

From the 3 axioms of probability, X and Y are independent if and only if, for all x, y,

$$\boldsymbol{F(x, y) = F_X(x)\, F_Y(y)}.$$

Equivalently, X and Y are independent if

$$p(x, y) = p_X(x)p_Y(y); \quad \text{discrete case,}$$

$$f(x, y) = f_X(x)f_Y(y); \quad \text{continuous case.}$$

Interpretation: Knowing the value of one random variable doesn't affect the distribution of the other.

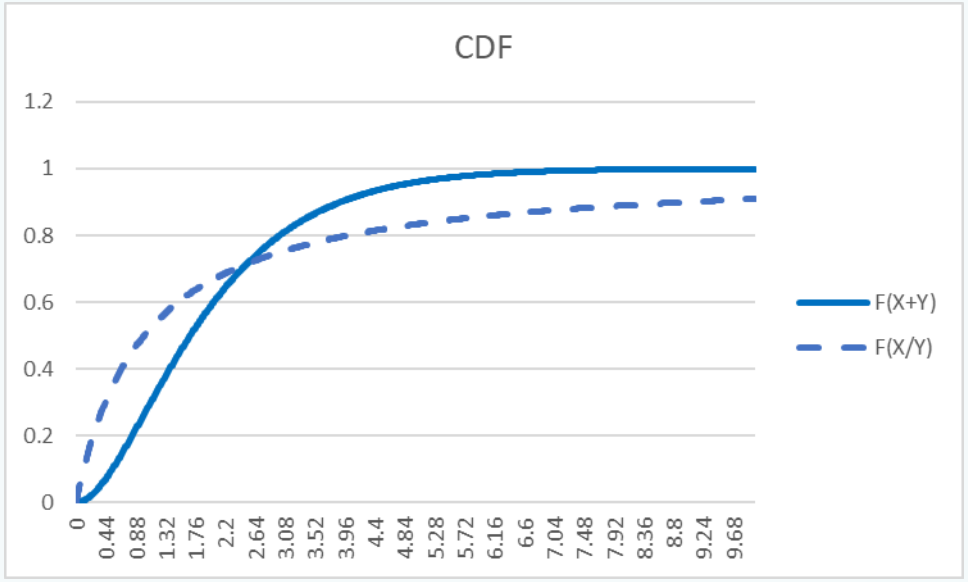
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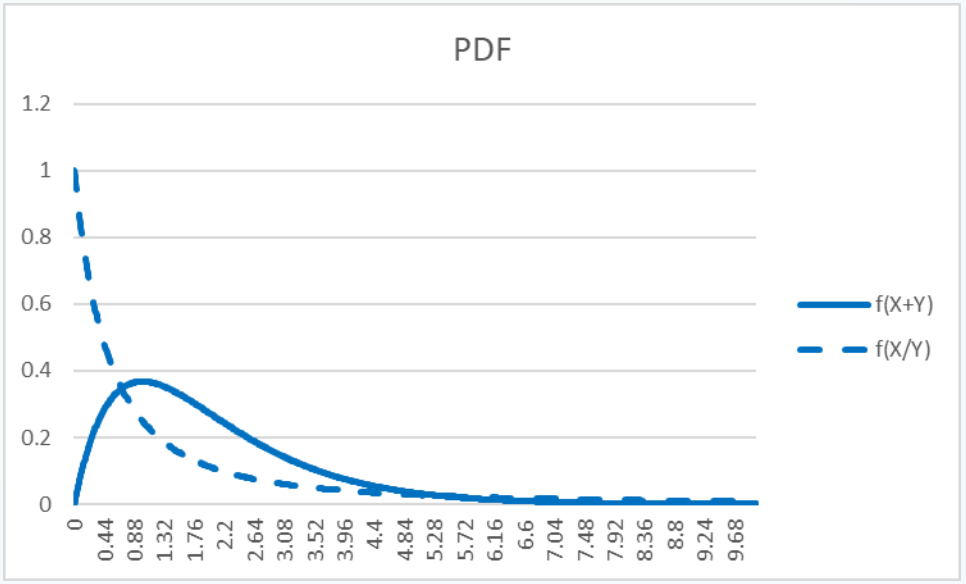
Example: X and Y are independent with pdf $f(x) = \begin{cases} e^{-x} & x > 0 \\ 0 & x \leq 0 \end{cases}$

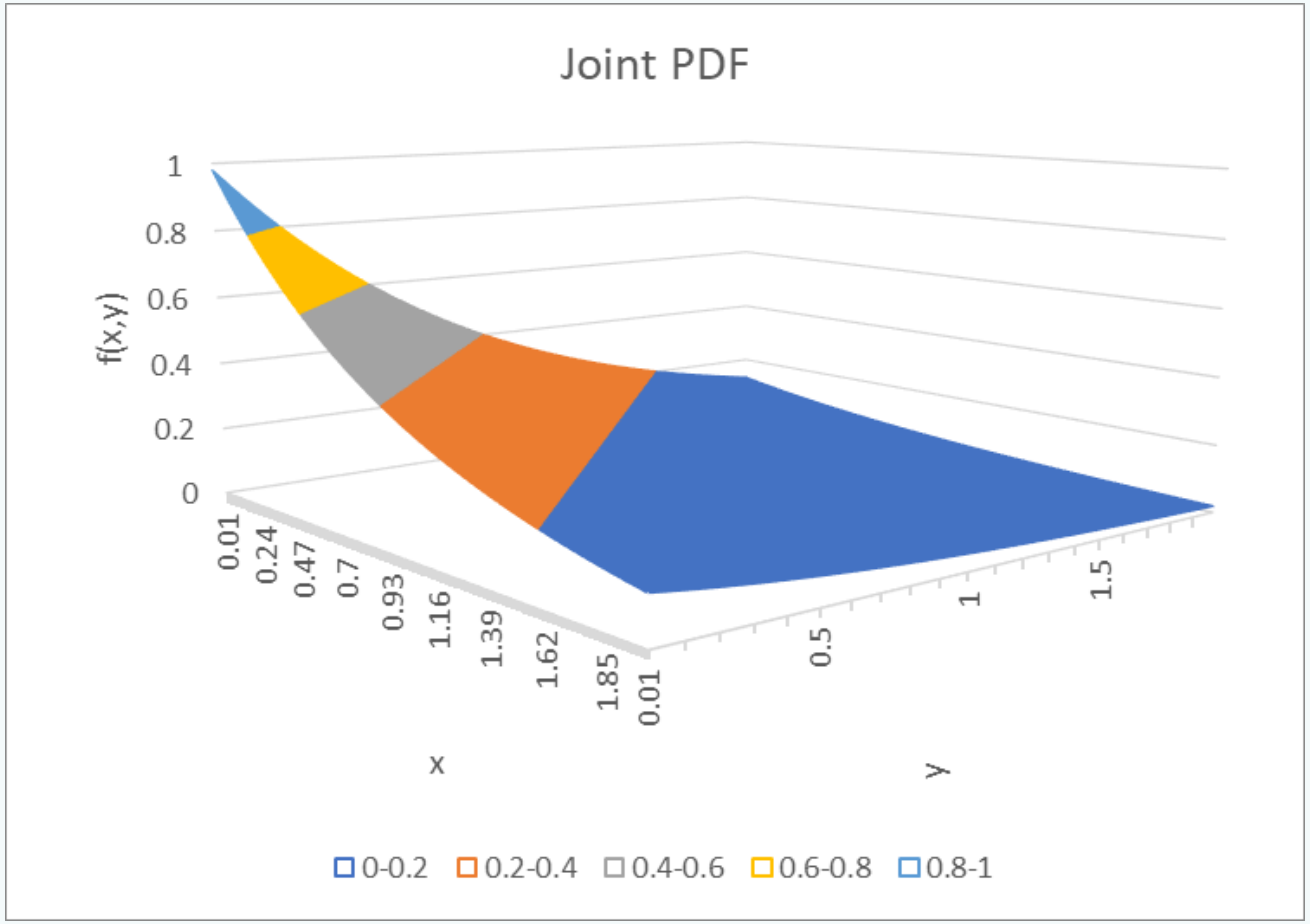
Find the CDF and the pdf of a) $X + Y$ b) X/Y .

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Example: X and Y are independent with pdf $f(x) = \begin{cases} e^{-x} & x > 0 \\ 0 & x \leq 0 \end{cases}$







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n random variables

The CDF of $X_1, X_2, \ldots, X_n$ is defined as

$$\boldsymbol{F(x_1, x_2, \ldots, x_n) = P\{X_1 \le x_1, X_2 \le x_2, \ldots, X_n \le x_n\}.}$$

If discrete, their joint probability mass function is

$$\boldsymbol{p(x_1, x_2, \ldots, x_n) = P\{X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n\}.}$$

They are jointly continuous if, for any set C, $f(x_1, x_2, \ldots, x_n)$ exists such that

$$\boldsymbol{P\{(X_1, X_2, \ldots, X_n) \in C\} = \iint_{(x_1, x_2, \ldots, x_n) \in C} f(x_1, x_2, \ldots, x_n)\, dx_1 dx_2 \ldots dx_n}$$

They are independent if, for any sets $A_i, i = 1,2,\ldots,n$,

$$P\{X_1 \in A_1, X_2 \in A_2, \ldots, X_n \in A_n\} = \prod_{i=1}^{n} P\{X_i \in A_i\}.$$

Or equivalently, if $F(x_1, x_2, \ldots, x_n) = \prod_{i=1}^{n} F_{X_i}(x_i)$.

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Example: Daily changes in the price of a stock are independent with common pmf

$$P(X = x) = \begin{cases} .05 & x = -2 \\ .20 & x = -1 \\ .35 & x = 0 \\ .25 & x = 1 \\ .10 & x = 2 \\ .05 & x = 3 \end{cases}$$

Find the probability that over the next three days, the price of the stock will increase in day 1, not increase in day 2 and increase by 2 in day 3.

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2.1.6 Conditional Distributions

The conditional probability mass function of X given that $Y = y$ is

$$p_{X|Y}(x|y) = P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{p(x, y)}{p_Y(y)}$$

for all y with $P_Y(y) > 0$.

Similarly, the conditional pdf of X given that $Y = y$ is

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}.$$

Since $f_{X|Y}$ is a valid pdf, conditional probabilities of events can be defined.

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Example: Let X and Y have the joint pdf

$$f(x,y) = \frac{12}{5}x(2-x-y), 0 < x < 1, 0 < y < 1$$

a) Find the conditional pdf of $X|Y$.

b) Find the probability that X is greater than 0.9 given that Y is i) 0.1, ii) 0.9.

a) $f_{X|Y}(x|y) =$

$f_Y(y) =$

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Example: Let X and Y have the joint pdf

$$f(x,y) = \frac{12}{5}x(2-x-y), 0 < x < 1, 0 < y < 1$$

a) Find the conditional pdf of $X|Y$.

b) Find the probability that X is greater than 0.9 given that Y is i) 0.1, ii) 0.9.

b) i) Y=.1

ii) Y=.9

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2.2 Moments of random variables

2.2.1 Expected Value

The expectation or expected value of a discrete random variable X is

$$\boldsymbol{E[X] = \sum_{i=1}^{\infty} x_i\, p(x_i)}\,.$$

Frequency interpretation:

For an infinite sequence of independent replications of a random experiment, the random variable X will take each possible value $x_i, i = 1,2, \ldots$ with a frequency (or proportion) equal to its probability, $\mathrm{p}(x_i)$.

The expectation or expected value of a continuous random variable X is

$$\boldsymbol{E[X] = \int_{-\infty}^{\infty} x\, f(x) dx}\,.$$

It is the weighted average of all possible values of X with "weights" $f(x)dx$.

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Remarks:

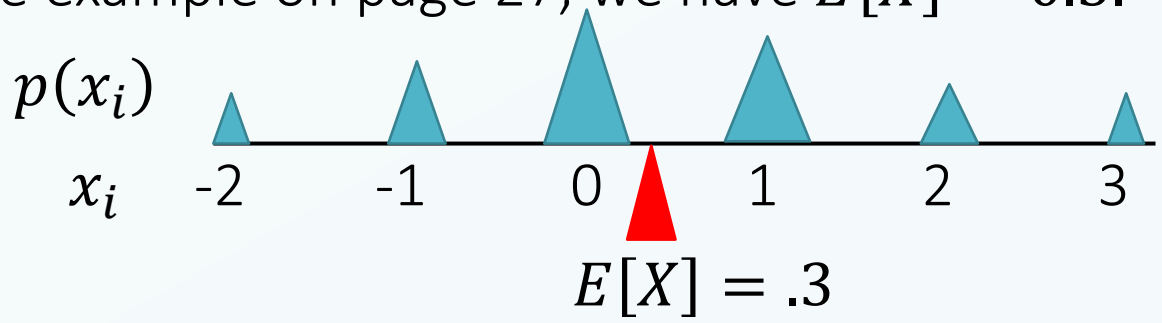
- The expected value doesn't have to be a possible value.

  For example, the expected value of the outcome when rolling a fair die is 3.5.

- The expected value is the center of gravity of the distribution.

  $E[X]$ is the point for which the total "torque" for all values of the distribution is 0.

  For the stock price example on page 27, we have $E[X] = 0.3$.



Total torque: $\sum_i (x_i - E[X])p(x_i)$ is set to 0.

- $E[X]$ is measured in the same units as X.

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Expected value of a function of a random variable

Suppose we are interested in the expected value of function of a random variable X.

Let Y be a random variable that is a function of X, say $Y = g(X)$.

We can obtain the expected value of Y by first finding its distribution.

Example: Let $f_X(x) = 1, 0 \le x \le 1$ and $Y = X^2$. Find $E[Y]$.

$$F_Y(y) = P\{Y \le y\} = P\{X^2 \le y\} = P\{0 \le X \le \sqrt{y}\}$$

$$= \int_0^{\sqrt{y}} f_X(x)dx = \int_0^{\sqrt{y}} 1\, dx = \sqrt{y} \qquad \text{for } 0 \le y \le 1.$$

$$f_Y(y) = \frac{d}{dy} y^{1/2} = \frac{1}{2} y^{-1/2} \qquad 0 \le y \le 1$$

And $E[Y] = \int_0^1 y \left(\frac{1}{2} y^{-1/2}\right) dy = \frac{1}{2}\frac{2}{3} y^{3/2} \Big|_0^1 = \frac{1}{3}.$

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Finding the distribution of $Y = g(X)$ can be challenging for some $g(X)$ and some distributions of X.

Another way to find the expected value of $Y = g(X)$ is to use the Law of the unconscious statistician.

For any real-valued function $g(x)$, the expected value of $\mathrm{Y} = g(X)$ is given by

$$E[Y] = \boldsymbol{E[g(X)]} = \begin{cases} \sum_{i=1}^{\infty} \boldsymbol{g(x_i)p(x_i)} & \textbf{if } \boldsymbol{X} \textbf{ is discrete} \\ \int_{-\infty}^{\infty} \boldsymbol{g(x)f_X(x)dx} & \textbf{if } \boldsymbol{X} \textbf{ is continuous} \end{cases}$$

Example: Let $f_X(x) = 1, 0 \leq x \leq 1$ and $Y = X^2$. Find $E[Y]$.

$$E[Y] = \int_{-\infty}^{\infty} g(x) f_X(x) dx = \int_0^1 x^2(1)\, dx = \frac{x^3}{3}\Big|_0^1 = \frac{1}{3}.$$

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Applying the Law of the unconscious statistician to the linear transformation,

$\mathrm{Y} = aX + b$,we get

$$E[Y] = \int_{-\infty}^{\infty} g(x) f_X(x) dx = \int_{-\infty}^{\infty} (ax + b) f_X(x) dx$$

$$= a \int_{-\infty}^{\infty} x f_X(x) dx \; + b \int_{-\infty}^{\infty} f_X(x) dx \; = a\, E[X] + b$$

The expected value is a linear operator.

The expected value is also called the mean or the first moment of X.

The nth moment of X for $n \geq 1$ is

$$\boldsymbol{E[X^n] = \begin{cases} \sum_{i=1}^{\infty} x_i^{\,n}\; p(x_i) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x^n\, f_X(x) dx & \text{if } X \text{ is continuous} \end{cases}}$$

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Expected value of sums of random variables

The two-dimensional version of the **Law of the unconscious statistician** states that

$$\boldsymbol{E[g(X,Y)]} = \begin{cases} \boldsymbol{\sum_{i=1}^{\infty}\sum_{j=1}^{\infty} g(x_i,y_j)p(x_i,y_j)} & \textbf{if } \boldsymbol{X, Y} \textbf{ are discrete} \\ \boldsymbol{\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y)f_{X,Y}(x,y)dx\,dy} & \textbf{if } \boldsymbol{X, Y} \textbf{ are jointly continuous} \end{cases}$$

Special case: Let $g(X,Y) = X + Y$ in the continuous case. Then,

$$E[g(X,Y)] = E[X+Y] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x+y)\,f(x,y)dx\,dy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x\,f(x,y)dx\,dy + \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} y\,f(x,y)dx\,dy$$

$$= \int_{-\infty}^{\infty} x\int_{-\infty}^{\infty} f(x,y)dy\,dx + \int_{-\infty}^{\infty} y\int_{-\infty}^{\infty} f(x,y)dx\,dy$$

$$= \int_{-\infty}^{\infty} x\,f_X(x)\,dx + \int_{-\infty}^{\infty} y f_Y(y)dy = \mathrm{E[X]} + \mathrm{E[Y]}$$

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In general, the expected value of a sum of random variables is the sum of their expected values.

$$E[X_1 + X_2 + \cdots + X_n] = E[X_1] + E[X_2] + \cdots + E[X_n].$$

Example: A firm bids on 3 jobs that would generate revenues of 10, 20 and 40 (thousand) dollars. The probability of winning the jobs are respectively .2, .8 and .3. Find the firm's expected revenue.

Let $X_i = \textbf{revenue for job } i$ and X the total revenue.

$E[X] = E[X_1] + E[X_2] + E[X_3] = 10(.2) + 20(.8) + 40(.3) = 30$ (i.e. \$30,000)

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Example: Letters for N clients have been placed randomly in N addressed envelopes. Find the expected number of letters placed in their correct envelope.

Let $X_i = \begin{cases} 1 & \text{ith letter in the correct envelope} \\ 0 & \text{otherwise} \end{cases}$

and X the number of letters placed in the correct envelope.

$$E[X] = E[X_1] + E[X_2] + \cdots + E[X_N]$$

Since $P\{X_i = 1\} = \frac{1}{N}$ and $E[X_i] = \frac{1}{N}$, $E[X] = N\ \frac{1}{N} = 1.$

So, on average one letter will end up in the correct envelope for any N.

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Example: Suppose one receives one coupon per day for 10 days. There are 20 different coupons, each equally likely to be received on each day. What is the expected number of different coupons that one will receive.

Let $X_i = \begin{cases} 1 & \text{no coupon of type i in the set of 10} \\ 0 & \text{otherwise (i.e. at least 1)} \end{cases}$

and X the number of different coupons in the set of 10.

$$E[X] = E[1 - X_1] + E[1 - X_2] + \cdots + E[1 - X_{20}]$$

$$= 20 - (E[X_1] + E[X_2] + \cdots + E[X_{20}])$$

Since $P\{X_i = 1\} = \left(\frac{19}{20}\right)^{10}$ *i=1,2,…,20;* and $E[X_i] = \left(\frac{19}{20}\right)^{10}$,

$$E[X] = 20 - 20\left(\frac{19}{20}\right)^{10} = 8.0253$$

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2.2.2 Median

The median of a continuous random variable is the value for which the CDf is 0.5. We will denote the median by m. It is determined by solving for $F(m) = 0.5$.

Example: Find the median of the random variables with pdf's

a) $f_X(x) = e^{-x}; \quad x > 0$

b) $f_Y(y) = 1; \quad 0 \leq y \leq 1$

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2.2.3 Best predictor of a random variable

Suppose you predict that the value of X will be c. The square of the prediction error is $(X-c)^2$ and the expected value is

$$\begin{aligned} E[(X-c)^2] &= E[(X-\mu+\mu-c)^2] \\ &= E[(X-\mu)^2 + 2(X-\mu)(\mu-c) + (\mu-c)^2] \\ &= E[(X-\mu)^2] + 2(\mu-c)E[X-\mu] + E[(\mu-c)^2] \\ &= E[(X-\mu)^2] + 2(\mu-c)(0) + (\mu-c)^2 \\ &\geq E[(X-\mu)^2] \end{aligned}$$

This implies that the smallest expected ***squared prediction error*** is achieved when $c = \mu$, that is predicting that the value of X will be its ***mean***.

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Suppose you predict that the value of X will be c. The absolute value of the prediction error is $|X - c|$ and the expected value is

$$E[|X - c|] = \int_{-\infty}^{\infty} |x - c| f(x) dx$$

$$= \int_{-\infty}^{c} (c - x)\, f(x) dx + \int_{c}^{\infty} (x - c)\, f(x) dx$$

$$= cF(c) - \int_{-\infty}^{c} x\, f(x) dx + \int_{c}^{\infty} x\, f(x) dx - c(1 - F(c))$$

We can find the value of c that minimizes the expected absolute prediction error by differentiating the expression with respect to c and setting it to 0.

$$\frac{d}{dc} E[|X - c|] = cf(c) + F(c) - cf(c) - cf(c) - \big(1 - F(c)\big) - c(-f(c))$$

$$= 2F(c) - 1 = 0 \quad \Rightarrow F(c) = 0.5 \Rightarrow c = m$$

This implies that the smallest expected ***absolute prediction error*** is achieved when predicting that the value of X will be its ***median***.

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2.2.4 Variance

The expected value of X summarizes the "location" of its distribution (i.e. the weighted possible values). The variation, or spread, is another key property of X. For example, the following random variables have the same mean, 0, but very different spreads.

$$X = 0 \quad w.p.1; \quad Y = \begin{cases} -1 & with\ probability\ 1/4 \\ 0 & w.p.\ 1/2 \\ 1 & w.p.1/4 \end{cases} ; \quad Z = \begin{cases} -100 & w.p.1/4 \\ 0 & w.p.\ 1/2 \\ 100 & w.p.1/4 \end{cases}$$

Since the distribution of X is centered around its mean, $\mathrm{E}[X] = \mu$, it makes sense to measure the dispersion by looking at how far the values are from the mean.

A possible measure would be the average distance from the mean, $\mathrm{E}[|X - \mu|]$. This measure of dispersion is mathematically inconvenient.

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Another measure of dispersion, more easily tractable, is the expectation of the squared distance between X and its mean.

Definition: The variance of X, a random variable with mean μ, is defined as

$$\boldsymbol{Var(X) = E[(X - \mu)^2]}$$

In practice, it may be easier to calculate the variance using

$$\boldsymbol{Var(X) = E[X^2] - \mu^2}$$

Exercise: Show that the above formula is equivalent to the definition of the variance.

$$Var(X) = E[(X - \mu)^2] = E[X^2 - 2\mu X + \mu^2]$$

$$= E[X^2] - E[2\mu X] + E[\mu^2]$$

$$= E[X^2] - \underline{\qquad\qquad} + \mu^2 =$$

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Example 1: Find the variance of an indicator variable. $I = \begin{cases} 1 & \text{if } A \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$.

Example 2: Let $\mathrm{f(x)} = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$. Find $Var(X)$.

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Example 3: Let $f(x) = \begin{cases} 2e^{-2x} & 0 < x \\ 0 & \text{otherwise} \end{cases}$. Find $Var(X)$.

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An identity

Let $Y = aX + b$, then for any constants a and b,

$$Var(Y) = Var(aX + b) = a^2\, Var(X)$$

Proof: Let $E[X] = \mu$, then $E[aX + b] = a\mu + b$. Using the definition of variance,

$$\begin{aligned}
\mathrm{Var}(aX + b) &= \mathrm{E}[(aX + b - E[aX + b])^2] \\
&= \mathrm{E}\left[\left(aX + b - (a\mu + b)\right)^2\right] = \mathrm{E}[(aX - a\mu)^2] \\
&= a^2\, E[(X - \mu)^2] = a^2\, Var(X).
\end{aligned}$$

Special cases: - $Var(aX) = a^2\, Var(X)$

- $Var(X + b) = Var(X)$

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Remarks:

-In physics, the variance represents the moment of inertia.

-Although the variance is the measure of dispersion defined here, in practice it is the standard deviation (the square root of the variance) that is the more meaningful measure.

-The standard deviation is measured in the same units as the values of X.

We denote the standard deviation of X by $\boldsymbol{\sigma_X}$.

$$\boldsymbol{\sigma_X = \sqrt{Var(X)}}$$

The standard deviation of a linear transformation, $Y = aX + b$, is

$$\boldsymbol{\sigma_Y = \sigma_{aX+b} = a\sigma_X}$$

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Example 1: The random variable X has expected value 10 and variance 25.

Find $E[(X+1)^2]$.

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Example 2: The time it takes to repair a device is a random variable X with pdf

$f(x) = 0.5;\ \ 0 < x < 2$. The cost depends on X and is $20 + 10\sqrt{x}$ if $X = x$.
Find the standard deviation of the cost.

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Example 3: Let $X_1, X_2, \ldots, X_5$ be independent random variables with common pdf

$$f(x) = 1, \quad 0 < x < 1 \text{ . Find } \mathrm{Var}(\max(X_1, X_2, \ldots, X_5)).$$

Let $\mathrm{X} = \max(X_1, X_2, \ldots, X_5)$. Using $\mathrm{E(X)} = \int_0^1 x f_X(x) dx$ and $\mathrm{E}(X^2) = \int_0^1 x^2 f_X(x) dx$ where $f_X(x)$ is

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2.2.5 Covariance of two random variables

The covariance of two random variables X and Y, denoted $\mathbf{Cov}(X, Y)$, is defined by

$$Cov(X, Y) = E\left[(X - \mu_x)(Y - \mu_y)\right].$$

It is an indicator of the strength of the (linear) relationship between the two random variables.

The correlation between X and Y is the dimensionless quantity that measures the strength of the relationship. It is,

$$Corr(X, Y) = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}.$$

Note: A positive covariance/correlation indicates that X and Y tend to move in the same direction (X *tends* to increase when Y increases and X *tends* to decrease when Y decreases). A negative covariance/correlation indicates that X and Y tend to move in opposite direction.

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A useful formula to calculate the covariance of X and Y is.

$$\boldsymbol{Cov(X,Y) = E[XY] - \mu_x\,\mu_y}$$

Proof: $Cov(X,Y) = E\big[(X-\mu_x)(Y-\mu_y)\big]$

$$= E\big[XY - \mu_x Y - \mu_y X + \mu_x\mu_y\big]$$

$$= E[XY] - \mu_x E[Y] - \mu_y E[X] + \mu_x\,\mu_y$$

$$= E[XY] - \mu_x\,\mu_y - \mu_y\,\mu_x + \mu_x\,\mu_y = E[XY] - \mu_x\,\mu_y.$$

Some basic *properties* of the covariance:

$$Cov(X,Y) = E[XY] - \mu_x\,\mu_y = E[YX] - \mu_y\mu_x = Cov(Y,X)$$

$$Cov(X,X) = E[X^2] - {\mu_x}^2 = Var(X)$$

$$Cov(aX,Y) = E[aXY] - (a\mu_x)\,\mu_y = a\big(E[XY] - \mu_x\,\mu_y\big) = aCov(X,Y),\ \forall a$$

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Additive property of the covariance:

$$Cov(X_1 + X_2, Y) = E[(X_1 + X_2)Y] - (\mu_{x_1} + \mu_{x_2})\mu_y$$
$$= E[(X_1Y + X_2Y)] - (\mu_{x_1}\mu_y + \mu_{x_2}\mu_y)$$
$$= E[X_1Y] - \mu_{x_1}\mu_y + E[X_2Y] - \mu_{x_2}\mu_y$$
$$= Cov(X_1, Y) + Cov(X_2, Y)$$

The additive property generalizes to

$$Cov(\sum_{i=1}^{n} X_i, Y) = \sum_{i=1}^{n} Cov(X_i, Y)$$

and

$$\boldsymbol{Cov\left(\sum_{i=1}^{n} X_i, \sum_{j=1}^{m} Y_j\right) = \sum_{i=1}^{n} Cov\left(X_i, \sum_{j=1}^{m} Y_j\right) = \sum_{i=1}^{n}\sum_{j=1}^{m} Cov(X_i, Y_j)}$$

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Variance of sums of random variables

Let $\mathrm{Y} = \sum_{i=1}^{n} X_i$. From the additive property of the covariance, the variance of a sum is given by

$$Var(\mathrm{Y}) = Cov(Y, Y) = Cov\left(\sum_{i=1}^{n} X_i \ , \sum_{j=1}^{n} X_j\right) = \boldsymbol{Var\left(\sum_{i=1}^{n} X_i\right)}$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{n} Cov(X_i, X_j)$$

$$= \sum_{i=1}^{n} Cov(X_i, X_i) + \sum_{i=1}^{n} \sum_{j \neq i} Cov(X_i, X_j)$$

$$= \boldsymbol{\sum_{i=1}^{n} Var(X_i) + \sum_{i=1}^{n} \sum_{j \neq i} Cov(X_i, X_j)}$$

Because $Cov(X_i, X_j) = Cov(X_j, X_i)$, we also have

$$\boldsymbol{Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i) + 2 \sum_{i=1}^{n} \sum_{j=i+1}^{n} Cov(X_i, X_j)}$$

For n=2, $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$

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Remarks:

1. For n independent random variables X_i, i=1,2,…,n, the variance of their sum is

$$\boldsymbol{Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i)}$$

since $Cov(X_i, X_j) = 0, \text{for all } i \neq j.$

2. The variance of a sum of random variables is the sum of all elements in the variance-covariance matrix, Σ, for these random variables.

$$\Sigma = \begin{pmatrix} Var(X_1) & Cov(X_1, X_2) & \cdots & Cov(X_1, X_{n-1}) & Cov(X_1, X_n) \\ Cov(X_2, X_1) & Var(X_2) & \cdots & Cov(X_2, X_{n-1}) & Cov(X_2, X_n) \\ \vdots & & \ddots & \vdots & \\ Cov(X_n, X_1) & Cov(X_n, X_2) & \cdots & Cov(X_n, X_{n-1}) & Var(X_n) \end{pmatrix}$$

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Example 1: Find $Var(X+Y)$ if their joint pdf is

$$f(x,y)=\begin{cases} e^{-x-y} & x>0, y>0 \\ 0 & \text{otherwise} \end{cases}$$

$$E(X)=\int_0^\infty \int_0^\infty x\ f(x,y)dydx=\int_0^\infty x\,e^{-x}\int_0^\infty e^{-y}dy\,dx=\int_0^\infty x\,e^{-x}dx=1$$

$$E(X^2)=\int_0^\infty x^2\,e^{-x}\int_0^\infty e^{-y}dy\,dx=\int_0^\infty x^2\,e^{-x}dx=2; \quad \text{and } Var(X)=1$$

Similarly, $E(Y)=1$ and $Var(Y)=1$

$$E(XY)=\int_0^\infty \int_0^\infty xy\ f(x,y)dydx=\int_0^\infty x\,e^{-x}\int_0^\infty ye^{-y}dy\,dx=1$$

$$\text{Cov}(X,Y)=E(XY)-E(X)E(Y)=0$$

$$Var(X+Y)=Var(X)+Var(Y)+2\text{Cov}(X,Y)=2$$

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Example 2: $X \; and \; Y$ have joint pdf given by

$$f(x,y) = \begin{cases} c(x+y) & 0 < x < k, 0 < y < x \\ 0 & \text{otherwise} \end{cases}$$

a) Find c
b) Find $\mathrm{E}(X)$ and $\mathrm{E}(\mathrm{Y})$
c) Find $\mathrm{Var}(X)$ and $\mathrm{Var}(\mathrm{Y})$
d) Find $\mathrm{Var}(X+Y)$

a)

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Example 2: X *and* Y have joint pdf given by

$$f(x,y) = \begin{cases} c(x+y) & 0 < x < k, 0 < y < x \\ 0 & \text{otherwise} \end{cases}$$

b) Find $\mathrm{E}(X)$ and $\mathrm{E}(\mathrm{Y})$

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Example 2: X *and* Y have joint pdf given by

$$f(x,y) = \begin{cases} c(x+y) & 0 < x < k, 0 < y < x \\ 0 & \text{otherwise} \end{cases}$$

c) Find $\mathrm{Var}(X)$ and $\mathrm{Var}(\mathrm{Y})$

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Example 2: $X \; and \; Y$ have joint pdf given by

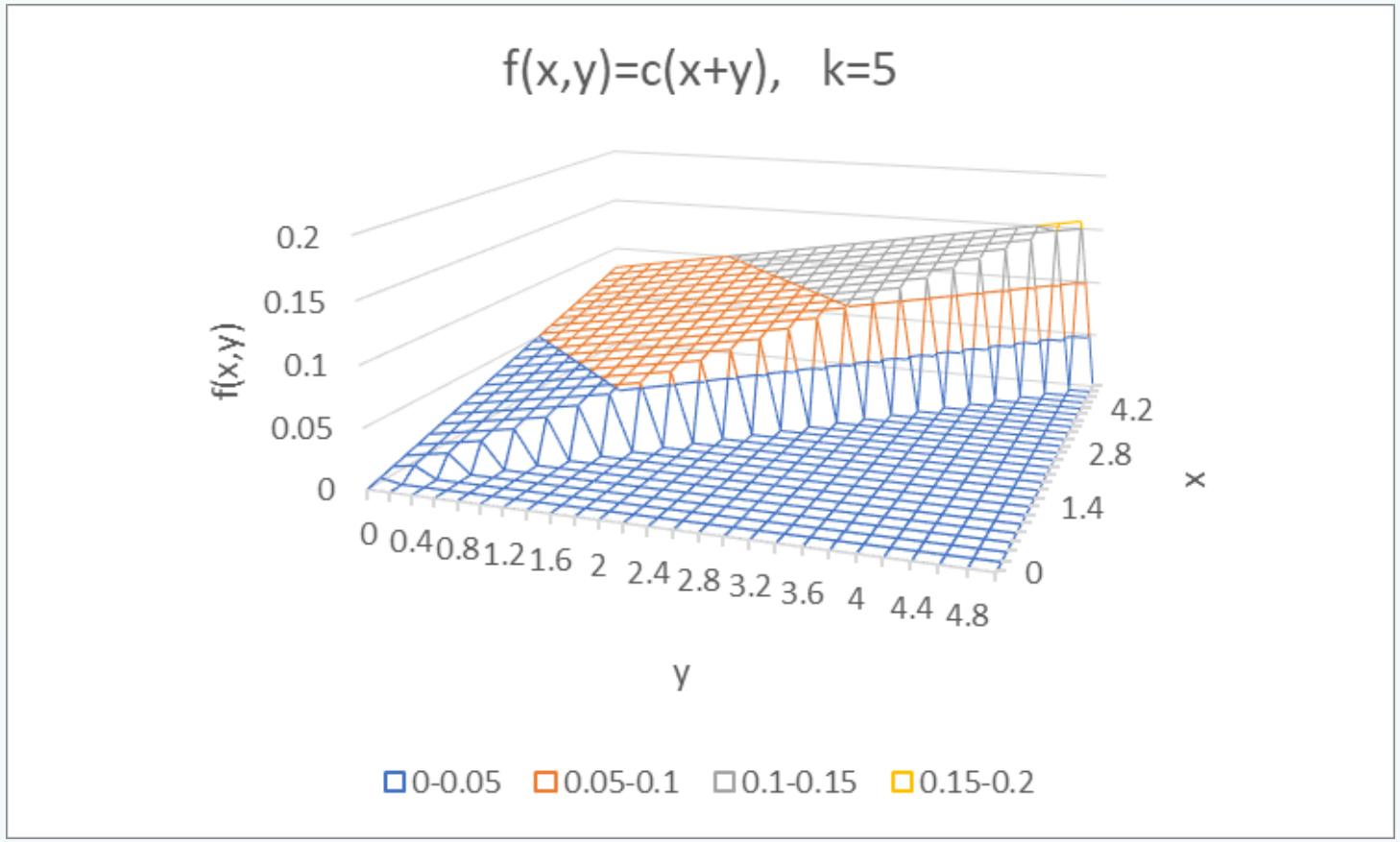
$$f(x,y) = \begin{cases} c(x+y) & 0 < x < k, 0 < y < x \\ 0 & \text{otherwise} \end{cases}$$

d) Find $\text{Var}(X+Y)$

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Example 2: Find $Var(X+Y)$ if their joint pdf is

$$f(x,y)=\begin{cases} c(x+y) & 0<x<k, 0<y<x \\ 0 & \text{otherwise} \end{cases}$$



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Example 3: You play a game where the net gain, W, is a function of the sum of two dice, where $\mathrm{W} = \begin{cases} 0 & \text{if the sum is } 2 \text{ or } 3 \\ 4 & \text{if the sum is } 10, 11 \text{ or } 12 \\ -1 & \text{otherwise} \end{cases}$.

If you play the game 10 times, what is the variance of your total gain?

Let $X_i = \begin{cases} 1 & \text{if } W = 4 \\ 0 & \text{otherwise} \end{cases}$ and $Y_i = \begin{cases} 1 & \text{if } W = -1 \\ 0 & \text{otherwise} \end{cases}$, $i = 1,2, \dots, 10$ for game i, i=1,2,…,10. Let $N_j, j = 1,2,3$ be the number of games resulting in a net gain of 0, 4 and -1 respectively. Then,

$$N_2 = \sum_{i=1}^{10} X_i \text{ and } N_3 = \sum_{i=1}^{10} Y_i \qquad N_1 = 10 - N_2 - N_3$$

The total net gain is $G = 4N_2 - N_3$.

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The expected value and variance of the total net gain are

$$E[G] = 4E[N_2] - E[N_3] = 4\sum_{i=1}^{10} E[X_i] - \sum_{i=1}^{10} E[Y_i]$$

$$= 4(10)\left(\frac{1}{6}\right) - 10\left(\frac{3}{4}\right) = -\frac{5}{6}$$

$$Var(G) = 16Var(N_2) + Var(N_3) - 2Cov(4N_2, N_3) \quad \text{where}$$

$$Var(N_2) = Var\left(\sum_{i=1}^{10} X_i\right) = 10\left(\frac{1}{6}\right)\left(\frac{5}{6}\right) = \frac{25}{18}$$

$$Var(N_3) = Var\left(\sum_{i=1}^{10} Y_i\right) = 10\left(\frac{3}{4}\right)\left(\frac{1}{4}\right) = \frac{15}{8}$$

$$Cov(N_2, N_3) = Cov\left(\sum_{i=1}^{10} X_i\,,\ \sum_{j=1}^{10} Y_j\right) = \sum_{i=1}^{10}\sum_{j=1}^{10} Cov(X_i, Y_j)$$

$$= \sum_{i=1}^{10} Cov(X_i, Y_i) + \sum_{i=1}^{10}\sum_{j\neq i} Cov(X_i, Y_j) \qquad Cov(X_i, Y_j) = 0$$

$$= \sum_{i=1}^{10} Cov(X_i, Y_i) = \sum_{i=1}^{10} E[X_iY_i] - E[X_i]E[Y_i] = 0 - 10\left(\frac{3}{4}\right)\left(\frac{1}{6}\right) = -\frac{5}{4}$$

$Var(G) = 22.22 + \frac{15}{8} + 8\left(\frac{5}{4}\right) = 34.097$ and the standard deviation is 5.839

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2.2.6 Moment Generating Functions

The moment generating function of X , denoted $\phi(t)$, is defined by

$$E[e^{tX}] = \begin{cases} \sum_x e^{tx}\ p(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} e^{tx}\ f(x)dx & \text{if } X \text{ is continuous} \end{cases}.$$

It is called the moment generating function (mgf) because the nth moment of X is equal to its nth derivative with respect to t, evaluated at t=0.

$$\phi^{(n)}(0) = E[X^n], \qquad \text{n} \geq 1$$

For example, $\phi'(t) = \frac{d}{dt}E[e^{tX}] = E\left[\frac{d}{dt}e^{tX}\right] = E[Xe^{tX}]$ and $\phi'(0) = E[X]$

$$\phi''(t) = \frac{d}{dt}\phi'(t) = E\left[\frac{d}{dt}Xe^{tX}\right] = E[X^2e^{tX}] \text{ and } \phi''(0) = E[X^2]$$

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Example: a) Find the mgf of X with pdf $f(x) = 1, \quad 0 < x < 1$.

b) Use the mgf to find the first two moments of X.

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A property of the mgf:

The mgf of a sum of independent random variables is the product of their mgf's.

Consider the independent random variables $X_i, i = 1,2, \dots, n$, then the mgf of their sum is

$$\phi_{\sum_{i=1}^{n} X_i}(t) = E\left[e^{t(X_1+X_2+\cdots+X_n)}\right] = E[e^{tX_1}e^{tX_2} \dots e^{tX_n}]$$

$$= E[e^{tX_1}]E[e^{tX_2}] \dots E[e^{tX_n}] \quad \text{by independence}$$

$$= \phi_{X_1}(t)\phi_{X_2}(t) \dots \phi_{X_n}(t) = \prod_{i=1}^{n} \phi_{X_i}(t)$$

Note: The *mgf uniquely determines the cdf* of the random variable.

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2.2.7 Conditional Moments

The conditional expected value of X given $Y = y$ is

$$E[X|Y = y] = \int_{-\infty}^{\infty} x \, f_{X|Y}(x|y)dx$$

This can be generalized to any arbitrary function. The expected value of $g(X, Y)$ given $Y = y$ is

$$E[g(X, Y)|Y = y] = \int_{-\infty}^{\infty} g(x, y) \, f_{X|Y}(x|y)dx$$

Recall that $f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{f_{X,Y}(x,y)}{\int_{-\infty}^{\infty} f_{X,Y}(x,y)dx}$

$$f_{X,Y}(x, y) = f_{X|Y}(x|y)f_Y(y) = f_{Y|X}(y|x)f_X(x)$$

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Since, in general, $E[X|Y=y]$ depends on y, replacing y by Y means that $E[X|Y]$ is also a random variable.

Now let $g(X,Y) = E[X|Y]$, then its expected value is

$$E[g(X,Y)] = E[E[X|Y]] = \int_{-\infty}^{\infty} E[X|Y=y]\, f_Y(y)dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x\, f_{X|Y}(x|y)dx\, f_Y(y)dy$$

$$= \int_{-\infty}^{\infty} x \int_{-\infty}^{\infty} f_{X|Y}(x|y)\, f_Y(y)dydx$$

$$= \int_{-\infty}^{\infty} x\, f_X(x)dx = E[X]$$

=> $\boldsymbol{E[E[X|Y]] = E[X]}$

This result is often referred to as the law of total expectation or the law of iterated expectations.

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The conditional variance of X given $Y = y$ is

$$Var[X|Y = y] = E[X^2|Y = y] - \{E[X|Y = y]\}^2$$

$$= \int_{-\infty}^{\infty} x^2 \, f_{X|Y}(x|y)dx - \{E[X|Y = y]\}^2$$

Now, $Var[X|Y]$ depends Y and is therefore a random variable. Its expected value is

$$E\big[Var[X|Y]\big] = E\big[E[X^2|Y]\big] - E[\{E[X|Y]\}^2] = E[X^2] - E[\{E[X|Y]\}^2]$$

Also, we can obtain the variance of the conditional expected value,

$$Var(E[X|Y]) = E[\{E[X|Y]\}^2] - \{E(E[X|Y])\}^2 = E[\{E[X|Y]\}^2] - E[X]^2$$

$$\boldsymbol{Var(X) = Var\big(E(X|Y)\big) + E(Var(X|Y))}$$

This result is often referred to as the variance decomposition formula or the law of total variance or the law of iterated variances.

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Example: Microcomputer chips are produced in two locations. Location A produces 80% of them and there is a 10% chance that anyone of them is defective. Those produced at location B have a 20% chance of being defective. They are sold in packages of 20 (all produced at the same location).

a) Find the expected number of defective chips in a package chosen at random.

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Example: Microcomputer chips are produced in two locations. Location A produces 80% of them and there is a 10% chance that anyone of them is defective. Those produced at location B have a 20% chance of being defective. They are sold in packages of 20 (all produced at the same location).

b) Find the variance of the number of defective chips in a package.

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2.3 Markov's inequality and Chebyshev's inequality

2.3.1 Expected value of a nonnegative random variable

Consider a nonnegative random variable X and assume its expected value exists. We can find its expectation by integrating the complement of its cdf. By definition,

$$E[X] = \int_0^\infty x\, f(x)dx$$

Integrating by parts, $u = x$ $dv = f(x)dx$

$$\mathrm{d}u = dx \qquad v = -(1 - F(x))$$

$$E[X] = -x\big(1 - F(x)\big)\Big|_0^\infty + \int_0^\infty (1 - F(x))dx$$

$$= 0 + \int_0^\infty (1 - F(x))dx \quad \text{since } E[X] < \infty$$

$$= \int_0^\infty (1 - F(x))dx$$

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Example 1: Find $\mathrm{E}[X]$ if $f(x) = ae^{-ax}, \quad 0 < x$.

$$\mathrm{F}(x) = 1 - e^{-ax}, \quad 0 < x$$

$$\mathrm{E}[X] = \int_0^\infty (1 - F(x))dx = \int_0^\infty e^{-ax}dx = -\frac{e^{-ax}}{a}\Big|_0^\infty = \frac{1}{a}$$

Example 2: Find $\mathrm{E}[X]$ if $f(x) = \begin{cases} 0.5 & 3 < x < 5 \\ 0 & \text{otherwise} \end{cases}$.

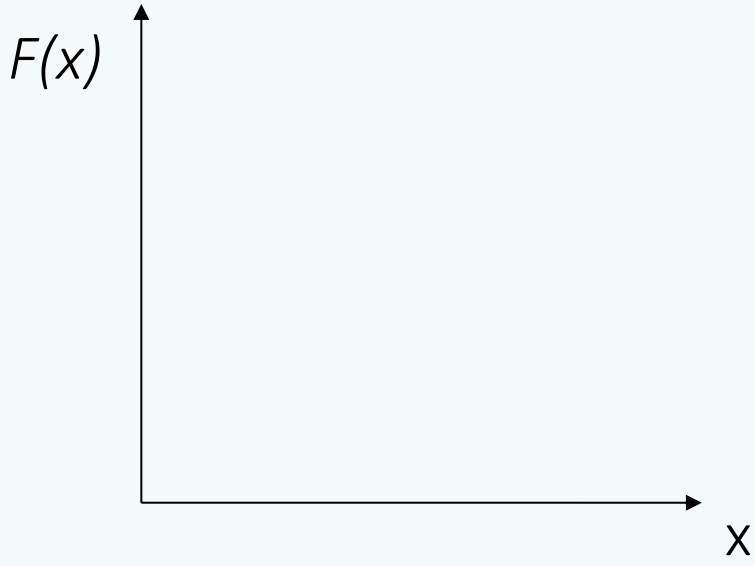
$$\mathrm{F}(x) = \begin{cases} 0 & x < 3 \\ 0.5(x-3) & 3 < x < 5 \\ 1 & 5 < x \end{cases} \longleftarrow \int_3^x 0.5dy = 0.5(x-3)$$

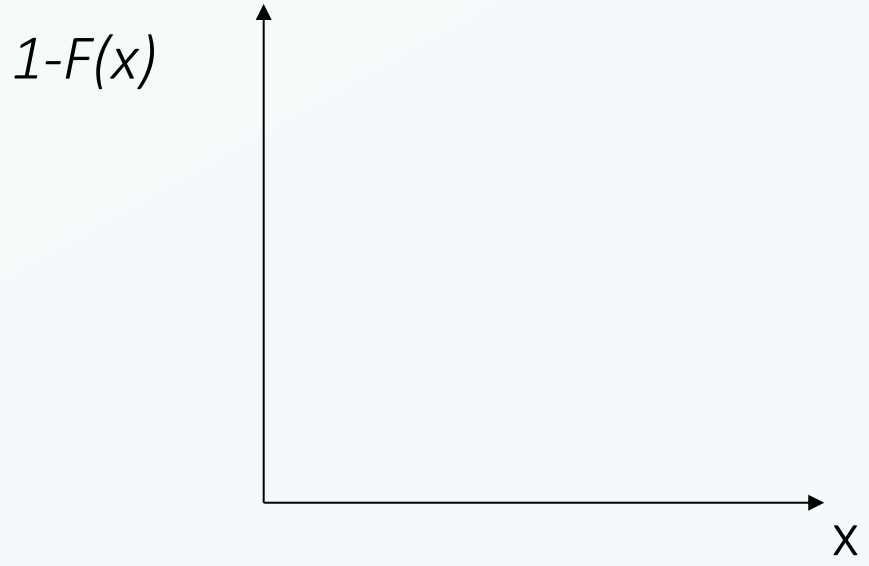
$$\mathrm{E}[X] = \int_0^\infty (1 - F(x))dx = \int_0^3 1\,dx + \int_3^5 0.5(5-x)dx + \int_5^\infty 0\,dx$$

$$= 3 + 0.5\left(5x - \frac{x^2}{2}\right)\Big|_3^5 = 3 + \frac{25}{4} - \frac{21}{4} = 4$$

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Example 2: Find $\mathrm{E}[X]$ if $f(x) = \begin{cases} 0.5 & 3 < x < 5 \\ 0 & \text{otherwise} \end{cases}$.





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2.3.2 Markov's inequality

Consider a nonnegative random variable X, then for any $a > 0$,

$$\boldsymbol{P\{X \geq a\} \leq \frac{E[X]}{a}}$$

Proof: $E[X] = \int_0^\infty (1 - F(x))dx$

$$= \int_0^a (1 - F(x))dx + \int_a^\infty (1 - F(x))dx$$

$$\geq \int_0^a \big(1 - F(x)\big)dx \geq \int_0^a \big(1 - F(a)\big)dx = a\big(1 - F(a)\big)$$

$$\Rightarrow \quad \frac{E[X]}{a} \geq \big(1 - F(a)\big) = \mathrm{P}\{\mathrm{X} \geq a\}$$

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2.3.3 Chebyshev's inequality

Consider a random variable X with mean μ and variance σ^2, then for any $\mathrm{k} > 0$,

$$\boldsymbol{P\{|X-\mu| \geq k\} \leq \frac{\sigma^2}{k^2}}$$

Proof: Applying Markov's inequality to the nonnegative random variable $(X-\mu)^2$,

$$P\{|X-\mu| \geq k\} = \mathrm{P}\{(X-\mu)^2 \geq k^2\} \leq \frac{E[(X-\mu)^2]}{k^2} = \frac{\sigma^2}{k^2}$$

Chebyshev's inequality is also written as

$$\boldsymbol{P\{|X-\mu| \geq k\sigma\} \leq \frac{1}{k^2}}$$

The probability that a random variable is more than k standard deviation away from its mean is less or equal to $\frac{1}{k^2}$.

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Example: The average number of items produced in a week is 50.

a) Find a bound for the probability that this week's production exceed 75.

b) The standard deviation of the weekly production is 5. Find a bound for the probability that this week's production will be between 40 and 60.

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2.4 Law of large numbers

Consider a sequence of independent and identically distributed random variables X_1, X_2,… with mean $E[X_i] = \mu$.

The (strong) law of large numbers states that that the sample average of n of these random variables, $\bar{X}_n = \frac{X_1+X_2+\cdots+X_n}{n}$, converges to μ with probability 1.

$$P\left\{\lim_{n\to\infty} \bar{X}_n = \mu\right\} = 1$$

Note: There is a (weak) law of large numbers.

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